

DMF Benchmarks - user manual

Configuring experiments and operating `dmf-benchctl`

DMF Benchmarks

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1. Purpose

This manual explains how to configure and operate DMF Benchmarks: how to prepare a configuration, validate it, run a benchmark, read the results, and publish an official run bundle.

The document is not tied to a single smoke dataset. Examples use concrete paths so they can be copied, but the flow is the same for LoCoMo, LongMemEval, DMF, and Mem0.

The system has two command-line surfaces with separate responsibilities. Users operate `dmf-benchctl` on the host. The controller manages the current orchestrator and explicitly starts `dmf-bench` inside the benchmark image. `dmf-bench` remains the scientific runtime; it does not start infrastructure itself.

All benchmarks are native-only and run inside Docker. The old protocol concept no longer exists.

2. Components

A standard execution uses these components:

- `dmf-benchctl`: the host-side control plane used by operators and automation.
- `benchmark`: the container that runs `dmf-bench`.
- `qdrant`: the local vector database used by the frameworks.
- `artifact-api`: a read-only API for reading runs and artifacts.
- `/bench/config`: read-only mounted configurations.
- `/bench/cache`: writable cache for materialized datasets, models, and temporary data.
- `/bench/runs`: writable volume where results are published.

Provider keys must never be written into JSON files. Pass them through environment variables, for example `OPENAI_API_KEY`.

3. Prerequisites

You need Docker with the Compose plugin, Python 3.12 or newer, the repository files, and the `dmf-benchctl` command installed from this project.

From the repository root, install the locked project package in your preferred Python environment:

```
python3 -m pip install .
```

Then check the two required host surfaces:

```
docker version
dmf-benchctl --help
```

If you use an image published to GHCR:

```
docker login ghcr.io
```

Create a local `.env` file in the repository root:

```
DMF_BENCH_IMAGE=ghcr.io/ORG/IMAGE:TAG
OPENAI_API_KEY=paste_the_key_here
HF_HUB_OFFLINE=1
```

`HF_HUB_OFFLINE=1` prevents implicit Hugging Face model downloads during an official run. If the cache does not already contain the required models, the run may fail; prepare the cache before the run.

No Qdrant API key is needed when Qdrant runs locally without authentication.

The `.env` file is read by the current Docker Compose adapter. You can also export the same variables in the shell. Never commit the file.

4. Configuration

An experiment is described by one JSON file with `schema_version: 2`.

The configuration is the scientific contract of a run. It says what benchmark is executed, which framework is evaluated, which dataset is used, which model answers, which model judges, where state is stored, and which reports must be produced.

A config should be treated as immutable once a run is considered official. If you change the dataset, framework config, selected units, model names, model parameters, or evaluation requirements, you are defining a different experiment.

The top-level structure is:

- `schema_version`: must be 2. Older benchmark config formats are rejected.
- `experiment_id`: stable human-readable experiment name. It should describe the benchmark, framework, and scope.
- `benchmark`: task to execute, either `locomot` or `longmemeval`.
- `framework`: memory framework under test, either `dmf` or `mem0`.
- `runtime`: operational paths and process settings. This includes root directory, run directory, cache directory, metrics port, and log level.
- `framework_config`: framework-specific TOML or YAML file plus SHA-256. This pins the DMF/Mem0 runtime configuration used by the experiment.
- `qdrant`: how the benchmark reaches Qdrant. The URL is read from an environment variable, normally `QDRANT_URL`.
- `dataset`: dataset declaration. It can point to an already materialized JSON file or to a registry entry that `dmf-bench` materializes before the run.
- `selection`: exact benchmark units to execute and optional filters. This controls the experiment size and order.
- `models`: provider, requested model, generation parameters, timeout, rate limit, and retry policy for answerer and judge.
- `evaluation`: reports that must or may be produced. Missing required reports make the run fail.
- `artifact_store`: destination for committed artifacts, normally the same directory as `runtime.runs_dir`.

The most important rule is separation of responsibilities:

- JSON config files define reproducible experiment inputs.
- Environment variables provide secrets and deployment endpoints.
- Run artifacts record what actually happened.

Do not put provider API keys in the JSON config. For example, `models.answerer.provider` may be `openai`, but the actual key must come from `OPENAI_API_KEY`. Similarly, `qdrant.endpoint_env` normally contains the string `QDRANT_URL`; the actual URL is read from that environment variable at runtime.

Path rules

When running inside the official Docker image, paths normally live under `/bench`:

`/bench/config` read-only configuration files
`/bench/cache` writable cache for datasets, models, and temporary data
`/bench/runs` writable output directory for committed run artifacts

Paths used by the benchmark must be absolute inside the runtime environment and must stay under `runtime.root`, normally `/bench`. This prevents a config from accidentally reading or writing unrelated host files.

In commands, `dmf-benchctl` accepts either a host path or an existing `/bench/...` path for the experiment config. Repository paths under `config/` and `smoke/` are translated automatically. A config elsewhere is mounted read-only at `/bench/operator/<filename>`. Paths written inside the JSON still describe the container runtime and therefore normally remain under `/bench`.

For example:

```
{
  "runtime": {
    "root": "/bench",
    "runs_dir": "/bench/runs",
    "cache_dir": "/bench/cache",
    "metrics_port": 9464,
    "log_level": "INFO"
  },
  "artifact_store": {
    "type": "local",
    "uri": "/bench/runs"
  }
}
```

Scientific versus operational fields

Some fields define the scientific identity of the run. Changing them changes the experiment:

- `benchmark`
- `framework`
- `framework_config.sha256`
- `dataset`
- `selection`
- `models.answerer.requested_model`
- `models.answerer.parameters`
- `models.judge.requested_model`
- `models.judge.parameters`
- `evaluation`

Other fields are operational. They can change where or how the same experiment is executed, but they should still be recorded for auditability:

- `runtime.runs_dir`
- `runtime.cache_dir`
- `runtime.metrics_port`
- `runtime.log_level`
- `qdrant.request_timeout_seconds`
- `artifact_store.uri`

`dmf-bench` records both kinds of information in the run manifest and provenance artifacts, but the scientific fingerprint focuses on the scientific inputs.

Minimal shape

A complete config has this shape:

```

{
  "schema_version": 2,
  "experiment_id": "locomo-dmf-official-run",
  "benchmark": "locomo",
  "framework": "dmf",
  "runtime": {
    "root": "/bench",
    "runs_dir": "/bench/runs",
    "cache_dir": "/bench/cache",
    "metrics_port": 9464,
    "log_level": "INFO"
  },
  "framework_config": {
    "path": "/bench/config/locomo_dmf_qdrant_settings.toml",
    "sha256": "64_character_sha256",
    "format": "toml",
    "profile": "official-v1"
  },
  "qdrant": {
    "endpoint_env": "QDRANT_URL",
    "retention": "keep",
    "request_timeout_seconds": 10
  },
  "dataset": {
    "name": "locomo",
    "registry_id": "locomo-official-unpinned",
    "allow_unpinned": true
  },
  "selection": {
    "ordered_item_ids": ["conversation-0001"],
    "filters": {},
    "seed": 7
  },
  "models": {
    "answerer": {
      "provider": "openai",
      "requested_model": "gpt-4.1-mini",
      "parameters": {
        "temperature": 0,
        "max_tokens": 4096
      },
      "runtime": {
        "timeout_seconds": 120,
        "rpm": 200,
        "max_retries": 5
      }
    },
    "judge": {
      "provider": "openai",
      "requested_model": "gpt-5-mini",
      "parameters": {
        "temperature": 0,
        "max_tokens": 4096,
        "reasoning_effort": "low"
      }
    }
  }
}

```

```

    "runtime": {
      "timeout_seconds": 120,
      "rpm": 200,
      "max_retries": 5
    }
  },
  "evaluation": {
    "required": ["primary_judge_score", "rigorous_report"],
    "optional": ["ablation_report"],
    "ablation_required": false
  },
  "artifact_store": {
    "type": "local",
    "uri": "/bench/runs"
  }
}

```

Before running an experiment, validate the config through the controller:

```
dmf-benchctl validate --materialize-datasets --config path/to/experiment.json
```

The controller mounts or translates the host path and runs `dmf-bench validate` inside the same image used for execution. Validation checks the config schema, path boundaries, framework config hash, dataset declaration, and environment-dependent settings that can be checked without executing the benchmark lifecycle.

5. Dataset

`dmf-bench` supports two ways to declare a dataset.

Already materialized dataset

Use this form when the dataset JSON file is already available:

```

{
  "dataset": {
    "name": "locomo",
    "source": "local-approved-dataset",
    "revision": "v1",
    "path": "/bench/datasets/locomo/locomo10.json",
    "sha256": "64_character_sha256",
    "expected_schema": "locomo10-json-array-v1"
  }
}

```

In this mode, `dmf-bench validate` checks that the file exists and that the SHA-256 matches.

Registry dataset

Use this form when you want `dmf-bench` to download or reuse a registered source:

```

{
  "dataset": {
    "name": "locomo",
    "registry_id": "locomo-official-unpinned",
    "allow_unpinned": true
  }
}

```

During run, the dataset is materialized under:

`/bench/cache/datasets/<fingerprint>/`

The directory always contains a `manifest.json`. The manifest documents:

- dataset id, benchmark, source URL, and revision;
- path and SHA-256 of the observed source file;
- path and SHA-256 of the materialized file;
- file sizes;
- optional sampling policy.

If the dataset is marked `unpinned`, `allow_unpinned: true` is required. This makes explicit that the run accepts a source that is not locked by an approved hash.

6. Sampling

Sampling is optional and is declared inside `dataset.sampling`.

LoCoMo example, 5% of QA stratified by category:

```
{
  "dataset": {
    "name": "locomo",
    "registry_id": "locomo-official-unpinned",
    "allow_unpinned": true,
    "sampling": {
      "fraction": 0.05,
      "unit": "qa",
      "stratify_by": "category",
      "seed": 7,
      "rounding": "ceil"
    }
  }
}
```

LongMemEval example, 5% of records:

```
{
  "dataset": {
    "name": "longmemeval",
    "registry_id": "longmemeval-s-official-unpinned",
    "allow_unpinned": true,
    "sampling": {
      "fraction": 0.05,
      "unit": "record",
      "seed": 7,
      "rounding": "ceil"
    }
  }
}
```

If `sampling` is omitted, `manifest.json` is still generated, but without a `sampling` block.

7. Selection

`selection.ordered_item_ids` decides which units are executed and in which order.

For LoCoMo, IDs are conversations, for example:

```
{
  "selection": {
    "ordered_item_ids": ["conversation-0001", "conversation-0002"],
    "filters": {"categories": [1, 2, 3, 4, 5]},
    "seed": 7
  }
}
```

For LongMemEval, IDs are `question_id` values, for example:

```
{
  "selection": {
    "ordered_item_ids": ["10d9b85a", "gpt4_45189cb4"],
    "filters": {},
    "seed": 7
  }
}
```

The selection must be consistent with the materialized dataset. If sampling is used, the selected IDs must exist in the generated sample.

8. Framework And Models

`framework_config` points to the framework configuration:

- DMF uses TOML files.
- Mem0 uses YAML files.

The file must provide `path`, `sha256`, `format`, and optionally `profile`.

`models.answerer` and `models.judge` define provider, model, and parameters. Example:

```
{
  "models": {
    "answerer": {
      "provider": "openai",
      "requested_model": "gpt-4.1-mini",
      "parameters": {"temperature": 0, "max_tokens": 4096},
      "runtime": {"timeout_seconds": 120, "rpm": 200, "max_retries": 5}
    },
    "judge": {
      "provider": "openai",
      "requested_model": "gpt-5-mini",
      "parameters": {
        "temperature": 0,
        "max_tokens": 4096,
        "reasoning_effort": "low"
      },
      "runtime": {"timeout_seconds": 120, "rpm": 200, "max_retries": 5}
    }
  }
}
```

Do not add `api_key`, `token`, `secret`, or similar fields to JSON files: validation rejects them.

9. Main Commands

User operations go through `dmf-benchctl`. Its current orchestrator adapter uses Docker Compose, but this detail is not part of the user contract. The controller always invokes `dmf-bench` explicitly for scientific runtime operations.

Show the controller commands:

```
dmf-benchctl --help
```

List benchmarks and frameworks:

```
dmf-benchctl runtime -- list
```

List registered datasets:

```
dmf-benchctl runtime -- list-datasets
```

Validate an already materialized configuration:

```
dmf-benchctl validate --config config/experiment.json
```

Validate after materializing the dataset from the registry:

```
dmf-benchctl validate --materialize-datasets --config config/experiment.json
```

Manually materialize a dataset:

```
dmf-benchctl runtime -- materialize-dataset \  
  --dataset-id locomo-official-unpinned \  
  --allow-unpinned \  
  --sample-fraction 0.05 \  
  --sample-unit qa \  
  --sample-stratify-by category \  
  --sample-seed 7 \  
  --sample-rounding ceil \  
  --output-dir /bench/cache/datasets/manual-locomotive-smoke
```

The command prints a `dataset` fragment usable in a configuration and the path to `manifest.json`.

The `runtime -- ...` escape hatch forwards advanced arguments to `dmf-bench` inside the image. Normal run, validation, state, verification, and publication operations have dedicated controller commands and should use those commands.

10. Execution

Before launching the run, set the image and provider variables required by the config, preferably in the repository `.env` file:

```
DMF_BENCH_IMAGE=ghcr.io/ORG/dmf-benchmarks:TAG  
OPENAI_API_KEY=...  
HF_HUB_OFFLINE=1
```

Do not set `QDRANT_URL` for the standard local stack. The benchmark container receives the internal endpoint `http://qdrant:6333` automatically.

Launch a run:

```
dmf-benchctl run --config config/experiment.json --run-id RUN_ID
```

`RUN_ID` must be unique and readable, for example:

```
official-20260809-locomotive-dmf
```


The controller starts Qdrant and the read-only Artifact API, leaves them running for result inspection, and executes `dmf-bench run` in a one-shot container. Add `--observability` to start Prometheus and Grafana as well. Add `--allow-downloads` only during an intentional cache-preparation or online run.

During the run, JSON events are printed. A completed run exits with code 0 and prints a final `COMPLETED` state. Stop services afterward without deleting run or cache volumes:

```
dmf-benchctl stack down
```

11. Artifact API

The Artifact API exposes results in read-only mode. When started locally, it normally responds at:

```
http://127.0.0.1:8000
```

Health check:

```
curl --fail http://127.0.0.1:8000/health
```

Run state:

```
curl --fail http://127.0.0.1:8000/runs/RUN_ID
```

Artifacts:

```
curl --fail http://127.0.0.1:8000/runs/RUN_ID/artifacts
```

The run state reports the final result and the local path to the `COMPLETED.json` marker:

```
curl --fail http://127.0.0.1:8000/runs/RUN_ID
```

To download a single artifact, use exactly one of the paths returned by `/runs/RUN_ID/artifacts`.

Cost and timing metrics:

```
curl --fail http://127.0.0.1:8000/runs/RUN_ID/artifacts/reports/usage.json
```

```
curl --fail http://127.0.0.1:8000/runs/RUN_ID/artifacts/reports/timing.json
```

12. Verification And Resume

Verify a published run:

```
dmf-benchctl verify --run-id RUN_ID
```

Inspect state:

```
dmf-benchctl status --run-id RUN_ID
```

Resume an interrupted run:

```
dmf-benchctl resume --run-id RUN_ID
```

Do not relaunch a run with the same `RUN_ID` before checking its state: you may duplicate provider calls.

13. Official Results

The official run result is published in the `/bench/runs` volume.

The most important files are:

Artifact	Content
<code>final/COMPLETED.json</code>	Final state, counts, and main references.
<code>final/manifest.json</code>	Run publication manifest.
<code>run-manifest.json</code>	Scientific run manifest.

Artifact	Content
<code>resolved-config.json</code>	Resolved configuration without secrets.
<code>reports/usage.json</code>	Tokens and provider usage when available.
<code>reports/timing.json</code>	Timing for measured phases and operations.
<code>reports/rigorous_report.json</code>	Deterministic rigorous metrics.
<code>reports/ablation_report.json</code>	Optional ablation report, if requested.
<code>logs/events.jsonl</code>	Lifecycle JSON events.
<code>datasets/materialization-manifest.json</code>	Dataset manifest copied into the run when the dataset is materialized from the registry.

To archive an official run, publish the run OCI bundle or keep at least the image name and digest, `RUN_ID`, `resolved-config.json`, dataset materialization manifest, `final/COMPLETED.json`, `reports/usage.json`, and `reports/timing.json`.

14. OCI Publication

A completed run can be published to GHCR as an OCI artifact. The artifact is not an executable image: it is a complete snapshot of the run directory, meaning everything under `runs/RUN_ID/`.

Before publication, the controller asks `dmf-bench` to verify the committed run and add these files to the run directory:

File	Purpose
<code>OFFICIAL-RUN.json</code>	Bundle metadata: OCI ref, subject, image, harness commit, and run verification.
<code>SHA256SUMS</code>	SHA-256 for every run file, excluding <code>SHA256SUMS</code> itself.

Command:

```
dmf-benchctl publish-run \
  --run-id RUN_ID \
  --ref ghcr.io/ORG/dmf-benchmarks-runs:RUN_ID \
  --subject ghcr.io/ORG/dmf-benchmarks@sha256:IMAGE_DIGEST
```

`--ref` is the GHCR destination for the run bundle. `--subject` links the bundle to the Docker image that produced the run.

Packaging happens inside the benchmark image against the shared run volume. The resulting archive is retained on the host, under `bundles/` by default. The controller then calls the host `oras` executable to push that exact archive. Authenticate `oras` to GHCR before publication.

To test packaging without publishing:

```
dmf-benchctl publish-run \
  --run-id RUN_ID \
  --ref ghcr.io/ORG/dmf-benchmarks-runs:RUN_ID \
  --output-dir bundles \
  --dry-run
```

A real push requires `oras` installed and authenticated to GHCR. A dry run does not require `oras` because it only verifies and exports the bundle.

15. Ready-To-Use Example

The `smoke/` directory contains ready-to-use configurations for a restricted but substantial check:

- LoCoMo with DMF: `smoke/config/experiment-locomo-dmf.json`
- LoCoMo with Mem0: `smoke/config/experiment-locomo-mem0.json`
- LongMemEval with DMF: `smoke/config/experiment-longmemeval-dmf.json`
- LongMemEval with Mem0: `smoke/config/experiment-longmemeval-mem0.json`

Run a ready config directly from the repository root:

```
dmf-benchctl run \  
--config smoke/config/experiment-locomo-dmf.json \  
--run-id smoke-20260809-locomo-dmf
```

The controller detects the `smoke/` path and automatically adds the required read-only mount. No orchestrator-specific option is needed.

This is only an operational example. For a real experiment, create a dedicated configuration and treat dataset, selection, models, and costs as explicit decisions.

16. Common Problems

Missing OPENAI_API_KEY. Check the host environment or repository `.env` file. Do not print the key in logs.

Config not found. For `dmf-benchctl --config`, verify the host path. For paths inside the JSON file, verify the corresponding `/bench` mount.

Dataset not found. If the config uses `registry_id`, use `run` or `validate --materialize-datasets`. If it uses `dataset.path`, check mounts and SHA-256.

SHA-256 mismatch. The file is not the one declared. Do not change the hash to make validation pass: regenerate or replace the file with the approved one.

Read-only file system. This is normal. Write only to `/bench/cache` and `/bench/runs`.

Qdrant not reachable. Run `dmf-benchctl stack status`. The standard stack injects `QDRANT_URL=http://qdrant:6333` automatically.

Download blocked. If `HF_HUB_OFFLINE=1`, Hugging Face downloads do not start. Prepare the cache before the run or change policy only for a controlled phase.

Provider rate limit or timeout. Check `logs/events.jsonl`, `reports/usage.json`, and run state before retrying.

OCI publication failed. Check that `oras` is installed, GHCR login is valid, and `dmf-benchctl verify --run-id RUN_ID` passes before pushing.

17. Operator Or Agent Checklist

1. Verify that Docker, Docker Compose, and `dmf-benchctl --help` respond.
2. Prepare `.env` without committing secrets.
3. Prepare or choose the JSON config.
4. Run `dmf-benchctl validate`; use `--materialize-datasets` if the config uses the registry.
5. Read the dataset `manifest.json` if the dataset was materialized.
6. Launch exactly one `dmf-benchctl run` with a unique `RUN_ID`; the controller starts required services.
7. Optionally request observability with `--observability`.
8. Check exit code and `COMPLETED` state.
9. Read artifacts from the Artifact API.
10. Save usage, timing, resolved config, and dataset manifest.
11. Publish the OCI bundle with `dmf-benchctl publish-run` if the run is official.

12. Stop services with `dmf-benchctl stack down`; named volumes are preserved.